Chapter A, Problem 2PP

Problem

Obtain the solution of this set of simultaneous equations:

$$\begin{aligned} 3x_1 - x_2 - 2x_3 &= 1 \\ -x_1 + 6x_2 - 3x_3 &= 0 \\ -2x_1 - 3x_2 + 6x_3 &= 6 \end{aligned}$$

Step-by-step solution

Step 1 of 4

Consider the following set of simultaneous equations,

$$\begin{aligned} 3x_1 - x_2 - 2x_3 &= 4 \\ -x_1 + 6x_2 - 3x_3 &= 0 \\ -2x_1 - 3x_2 + 6x_3 &= 6 \end{aligned}$$

Arrange the equations in matrix form.

$$\underbrace{\begin{bmatrix} 3 & -1 & -2 \\ -1 & 6 & -3 \\ -2 & -3 & 6 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}_{\mathbf{X}} = \underbrace{\begin{bmatrix} 1 \\ 0 \\ 6 \end{bmatrix}}_{\mathbf{B}}$$

Calculate the determinant of matrix **A**, Δ, by repeating the first two rows of the matrix.

$$\begin{aligned} \Delta &= \begin{vmatrix} 3 & -1 & -2 \\ -1 & 6 & -3 \\ -2 & -3 & 6 \end{vmatrix} \\ &= \begin{vmatrix} 3 & -1 & -2 \\ -1 & 6 & -3 \\ -2 & -3 & 6 \\ 3 & -1 & -2 \\ -1 & 6 & -3 \end{vmatrix} \\ &= \begin{bmatrix} (3)(6)(6)+(-1)(-3)(-2)+(-2)(-1)(-3)- \\ (-2)(6)(-2)-(3)(-3)(-3)-(-1)(-1)(6) \end{bmatrix} \\ &= 108-6-6-24-27-6 \\ &= 39 \end{aligned}$$

Step 2 of 4

Calculate the determinant Δ_1, which is the determinant of the matrix formed by replacing first column of matrix **A** by column matrix **B** .

$$\Delta_1 = \begin{vmatrix} 1 & -1 & -2 \\ 0 & 6 & -3 \\ 6 & -3 & 6 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & -1 & -2 \\ 0 & 6 & -3 \\ 6 & -3 & 6 \\ 1 & -1 & -2 \\ 0 & 6 & -3 \end{vmatrix}$$

$$= \begin{bmatrix} (1)(6)(6)+(0)(-3)(-2)+(6)(-1)(-3)- \\ \quad (6)(6)(-2)-(1)(-3)(-3)-(0)(-1)(6) \end{bmatrix}$$

$$= 36+0+18+72-9+0$$

$$= 117$$

Step 3 of 4

Calculate the determinant Δ_2, which is the determinant of the matrix formed by replacing second column of matrix **A** by column matrix **B** .

$$\Delta_2 = \begin{vmatrix} 3 & 1 & -2 \\ -1 & 0 & -3 \\ -2 & 6 & 6 \end{vmatrix}$$

$$= \begin{vmatrix} 3 & 1 & -2 \\ -1 & 0 & -3 \\ -2 & 6 & 6 \\ 3 & 1 & -2 \\ -1 & 0 & -3 \end{vmatrix}$$

$$= \begin{bmatrix} (3)(0)(6)+(-1)(6)(-2)+(-2)(1)(-3)- \\ \quad (-2)(0)(-2)-(3)(6)(-3)-(-1)(1)(6) \end{bmatrix}$$

$$= 0+12+6+0+54+6$$

$$= 78$$

Step 4 of 4

Calculate the determinant Δ_3, which is the determinant of the matrix formed by replacing third column of matrix **A** by column matrix **B** .

$$\Delta_3 = \begin{vmatrix} 3 & -1 & 1 \\ -1 & 6 & 0 \\ -2 & -3 & 6 \end{vmatrix}$$

$$= \begin{vmatrix} 3 & -1 & 1 \\ -1 & 6 & 0 \\ -2 & -3 & 6 \\ 3 & -1 & 1 \\ -1 & 6 & 0 \end{vmatrix}$$

$$= \begin{bmatrix} (3)(6)(6)+(-1)(-3)(1)+(-2)(-1)(0)- \\ \qquad (-2)(6)(1)-(3)(-3)(0)-(-1)(-1)(6) \end{bmatrix}$$

$$= 108+3+0+12+0-6$$
$$= 117$$

Determine x_1 by using Cramer's rule.

$$x_1 = \frac{\Delta_1}{\Delta}$$
$$= \frac{117}{39}$$
$$= 3$$

Determine x_2 by using Cramer's rule.

$$x_2 = \frac{\Delta_2}{\Delta}$$
$$= \frac{78}{39}$$
$$= 2$$

Determine x_3 by using Cramer's rule.

$$x_3 = \frac{\Delta_3}{\Delta}$$
$$= \frac{117}{39}$$
$$= 3$$

Thus, the solution to the simultaneous equations is $\boxed{x_1 = 3,\ x_2 = 2,\ x_3 = 3}$.